

Fat Soluble Vitamins (3 Parts)

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Vitamin A (part 1); Vitamin D (part 2);

Vitamin K and E (part 3)

Self-assessment questions in the videos



Vitamin A

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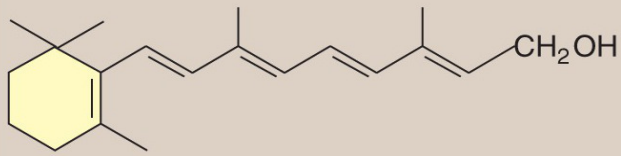
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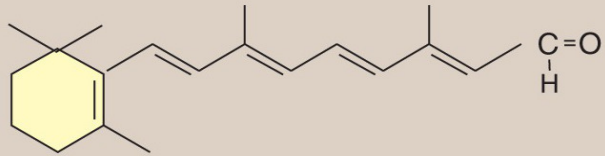
Objectives: DLA: Fat soluble vitamins (A, D, E and K)

- Before DM Lecture 33

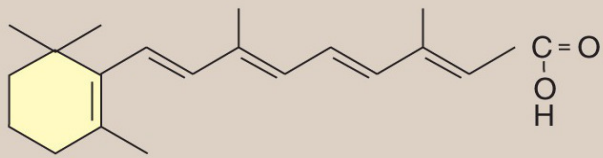
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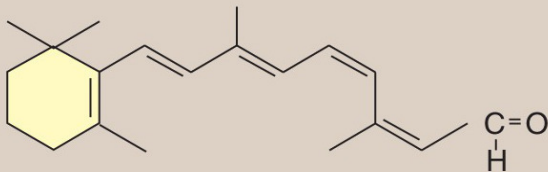
Retinol



Retinal



Retinoic acid
(all trans)



11-cis Retinal

Required for vision

VITAMIN A

- Collectively called retinoids
- Retinol – transport and storage form
- **11-cis retinal** is required for vision
- Retinol and retinal can be easily interconverted
- **Retinoic acid cannot be converted to retinol or retinal**
- Retinoic acid – epithelial growth and differentiation. It has steroid hormone like effects

Absorption and transport of Vitamin A

Diet contains retinol esters and beta-carotene



Absorption requires **bile salts**. Beta-carotene is converted to retinol



Esterification and secretion in **chylomicrons** as retinol esters taken up by the **liver** where it is stored



Plasma retinol binding protein (Transthyretin) transports vitamin A to extra hepatic tissues



Tissues contain cellular retinol binding protein that carries retinol into the cell

Sources and Functions of Vitamin A

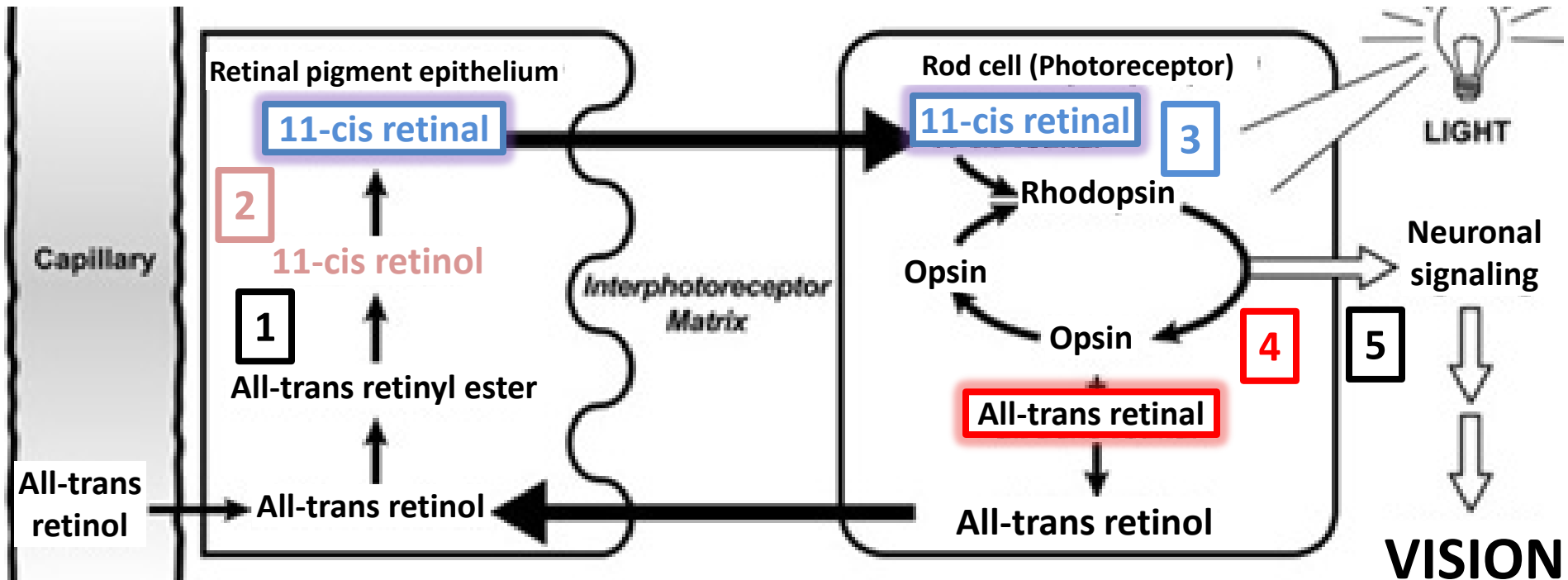
Dietary Sources

- Liver, kidney, butter and cream products; egg yolk
- Yellow vegetables and fruits (Dietary provitamin form: **β-carotene**)

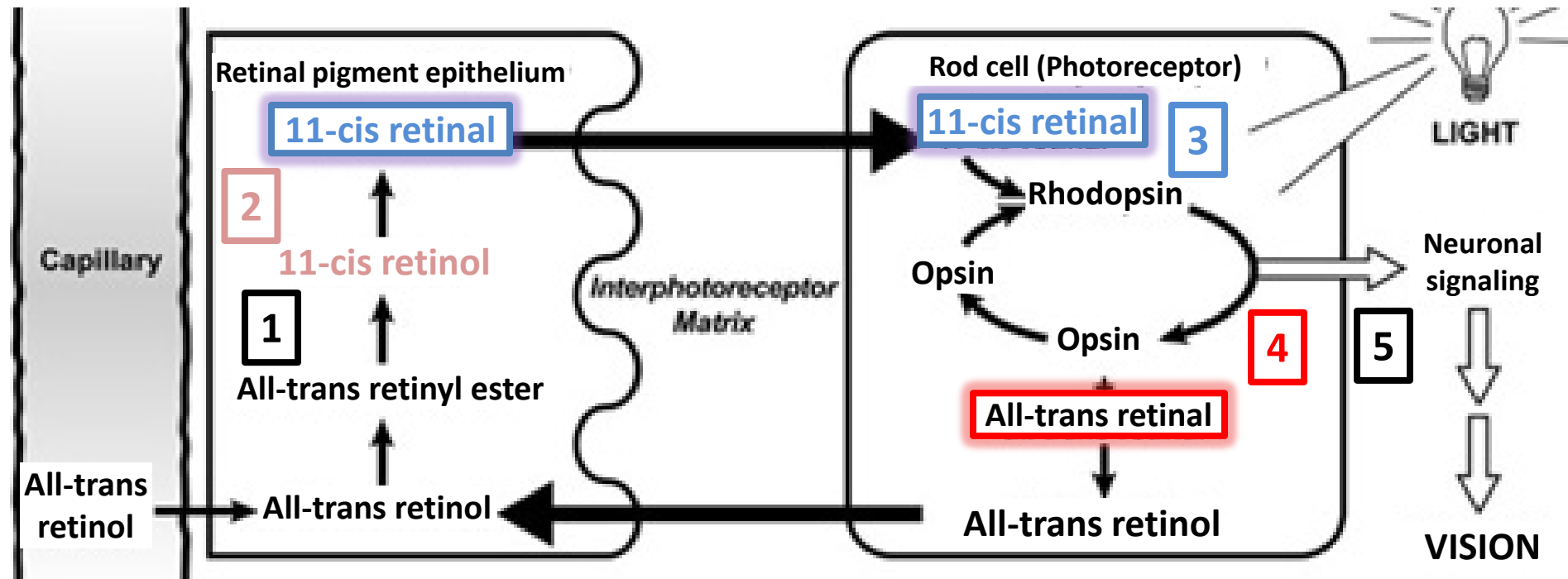
Functions

- **Vision** – **11 cis retinal** is a component of rhodopsin (visual pigment). Visual cycle requires the isomerization reaction between cis and trans-retinal.
- **Maintenance of specialized epithelia, especially mucus secreting cells (retinoic acid)**
- **Growth (retinoic acid)**
- **Reproduction (retinol)**

Role of 11-cis retinal in vision



Role of 11-cis retinal



1. All trans-retinol is isomerized in the retina to **11-cis retinol**
2. **11-cis retinol** is oxidized to **11-cis retinal**
3. **11-cis retinal** enters the rod cell where it combines with opsin to form **rhodopsin** (visual pigment)
4. Absorption of a photon of light catalyzes the isomerization of **11-cis-retinal** to **all-trans-retinal** triggering a cascade of events.
5. Leading to the generation of an electrical signal to the optic nerve which is interpreted as vision

Mechanism of action of retinoic acid in epithelial cells

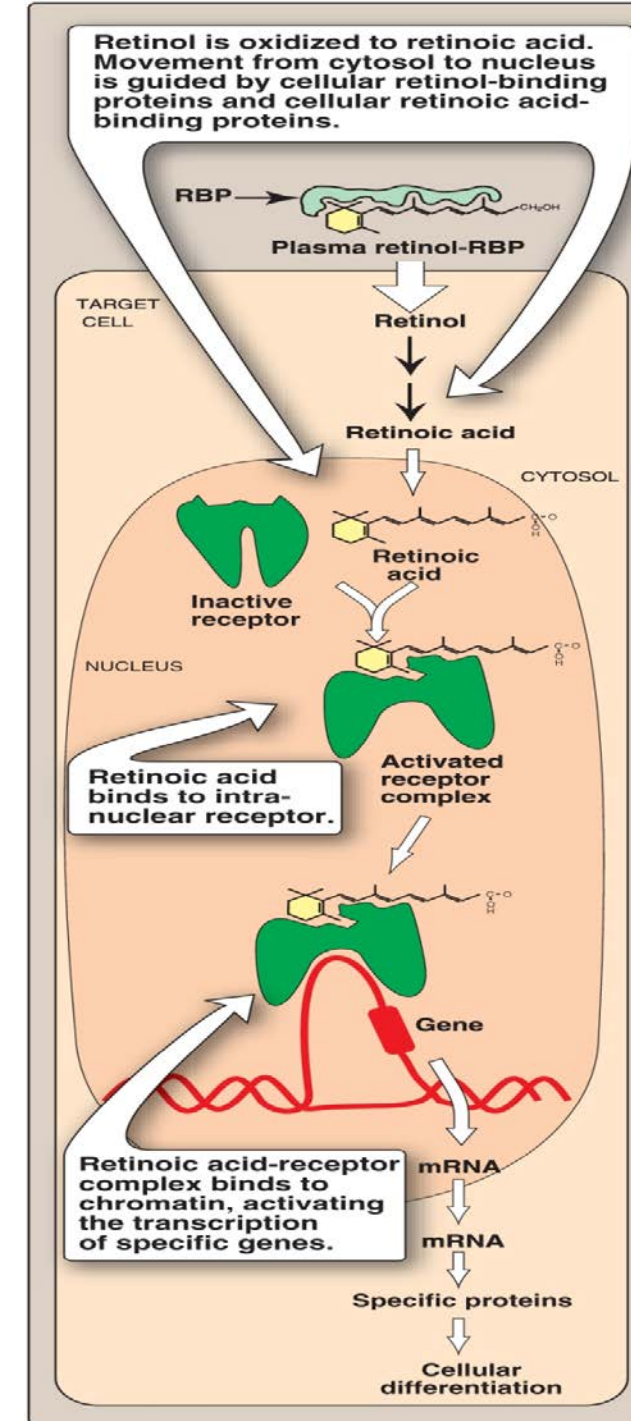
Retinol enters the target cell and is oxidized to **retinoic acid** in the cytosol

From the cytosol, the retinoic acid moves into the nucleus with the help of cellular retinoid binding proteins

Retinoic acid binds to **nuclear receptors** forming an activated receptor complex (acts like a **transcriptional regulator**)

Similar to mechanism of action of a **steroid hormone**

Retinoic acid-receptor complex binds to chromatin activating the **transcription of specific genes (keratin)**



Deficiency of Vitamin A

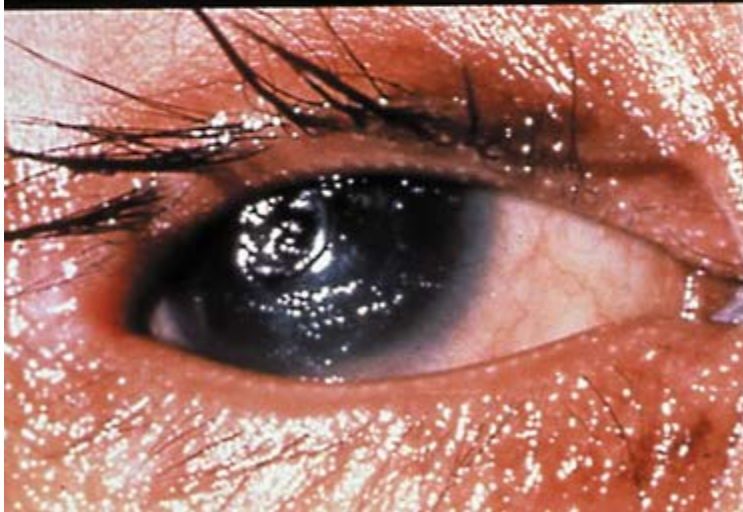
- Risk factors:
 - Dietary deficiency is the commonest cause – fat free diets; lack of dietary sources
 - Fad diets: [Teen Eats Chicken Nuggets for 15 Years, Nothing Else](#)
 - Malabsorption of fats (cystic fibrosis, obstructive jaundice)

Signs and symptoms

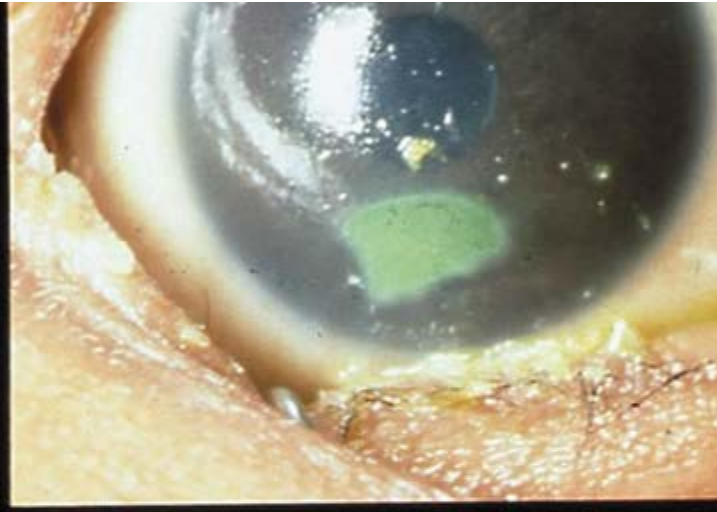
- Night blindness – earliest symptom; inability to see in dim light; Regeneration of rhodopsin is delayed
- Xerophthalmia – dryness of the conjunctiva and cornea.
- Bitot spots
- Keratomalacia – corneal erosion and ulceration
- Increased risk of pulmonary infections
- Immune deficiency – weakened innate immunity



Bitot spots



Keratomalacia



Corneal opacity

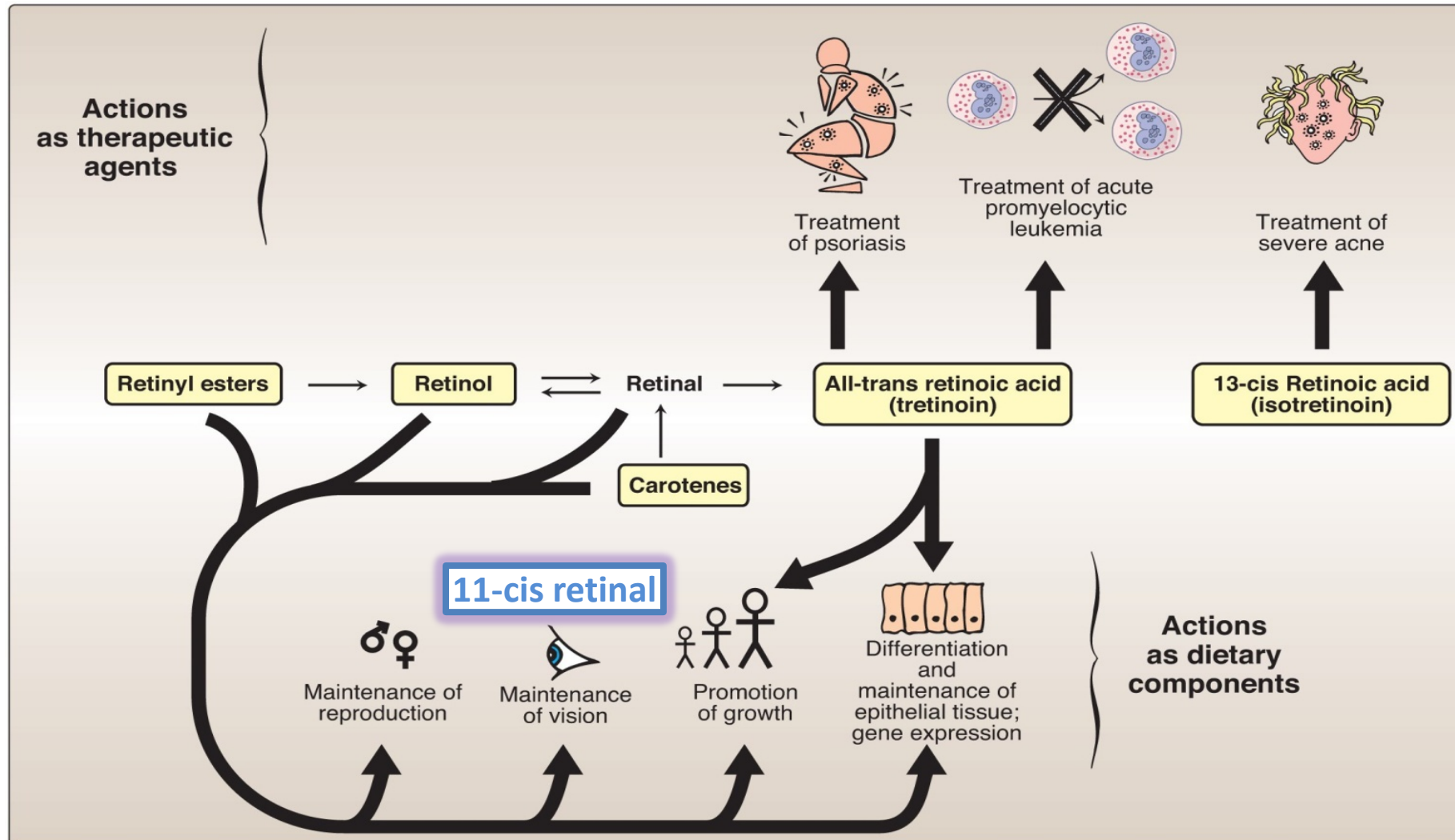
What are the dietary sources you would recommend for the children in the family to prevent the micronutrient deficiency developing in the other children?



Bitot spots

Deficiencies of dietary retinoids leads to night blindness, less fertility, slowed growth and skin problems.

Retinoic acid is used to treat severe acne and psoriasis.



Clinical uses

- Retinol or β -carotene: Treatment of dietary deficiency
- β -carotene is a **lipid soluble antioxidant (along with vitamin E)**
- Retinoic acid is used in the treatment of acne, especially topical treatment
- The all-trans retinoic acid is used in the treatment of acute promyelocytic leukemia

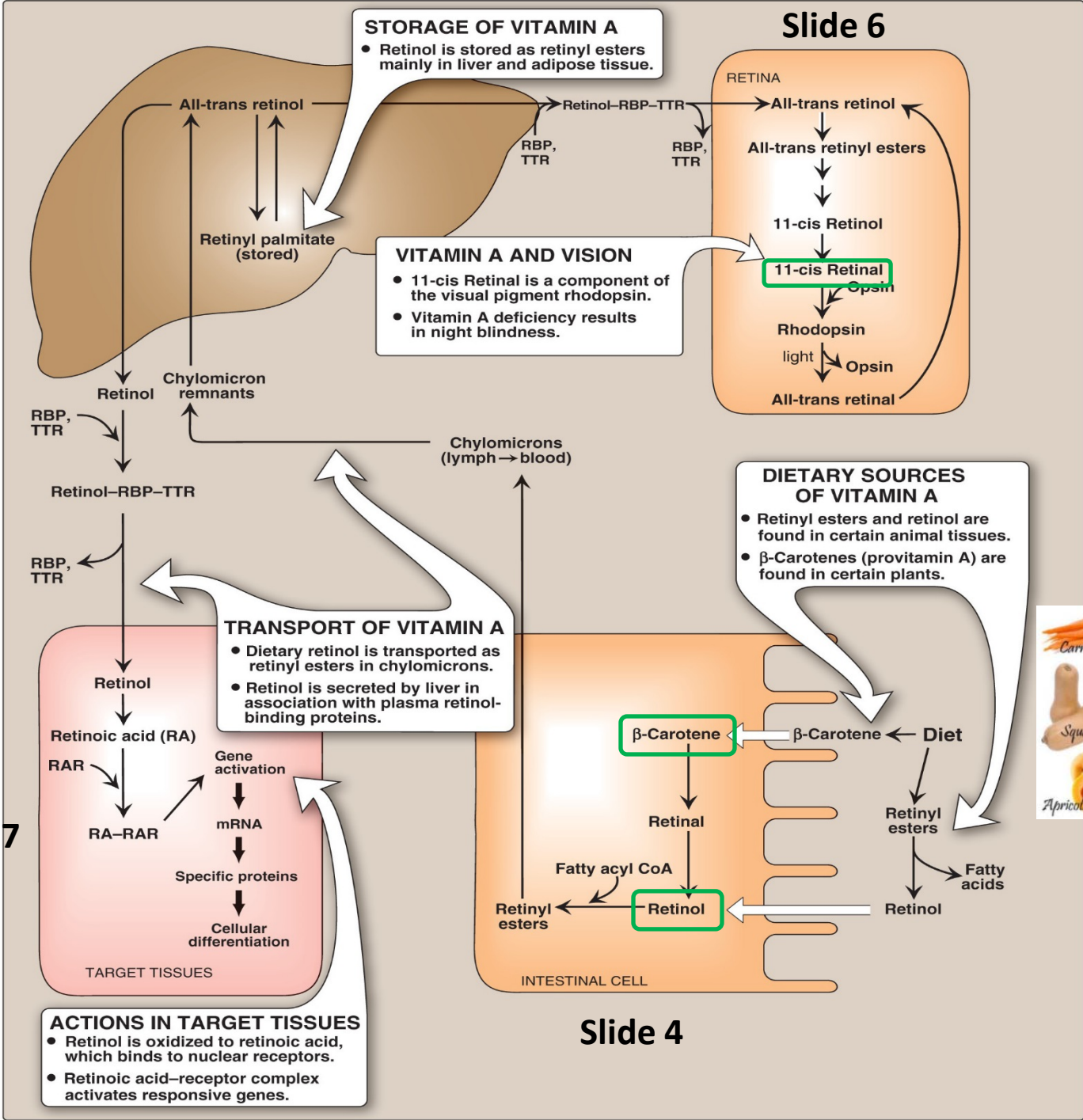
Hypervitaminosis A

- Dry and pruritic (itchy) skin – skin peeling
- Central nervous system – raised intracranial pressure – **headaches**
– may mimic brain tumors – **Benign intracranial hypertension**
- Enlarged liver and altered liver function tests
- <http://www.theguardian.com/science/antarctica-live/2013/dec/04/douglas-mawson-antarctic-trek>
- Vitamin A in Pregnancy – spontaneous abortions and congenital malformations in the fetus. (vitamin A is a teratogen)
 - ***Vitamin A supplements (greater than the 10,000 IU) should be avoided in pregnant women***

Summary: Vitamin A

Slide 4

Slide 7



SUNLIGHT
The body itself makes
vitamin D when it is
exposed to the sun

10 Best Vitamin D Rich Foods



Fortified Cereals



Orange Juice



Egg Yolk



Mushroom



Ricotta Cheese



Fatty Fish



COD Liver Oil



Cavier (Fish eggs)

Vitamin D:
‘The sunshine vitamin’

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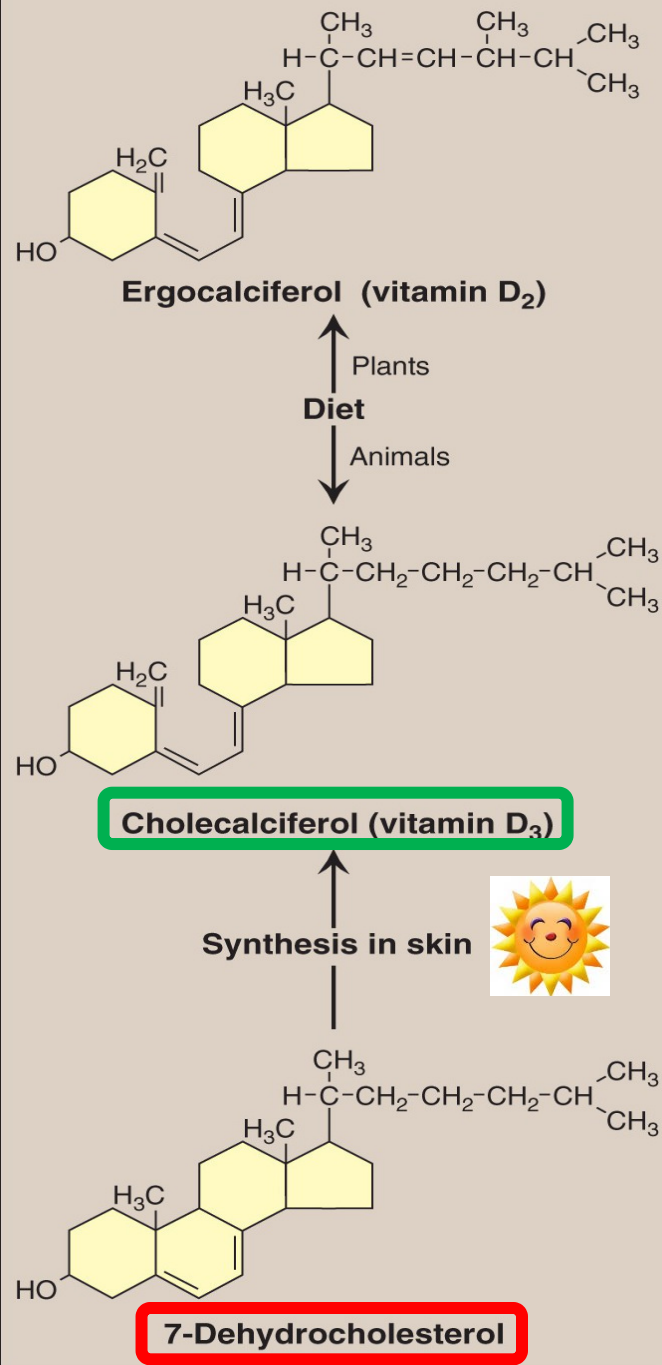
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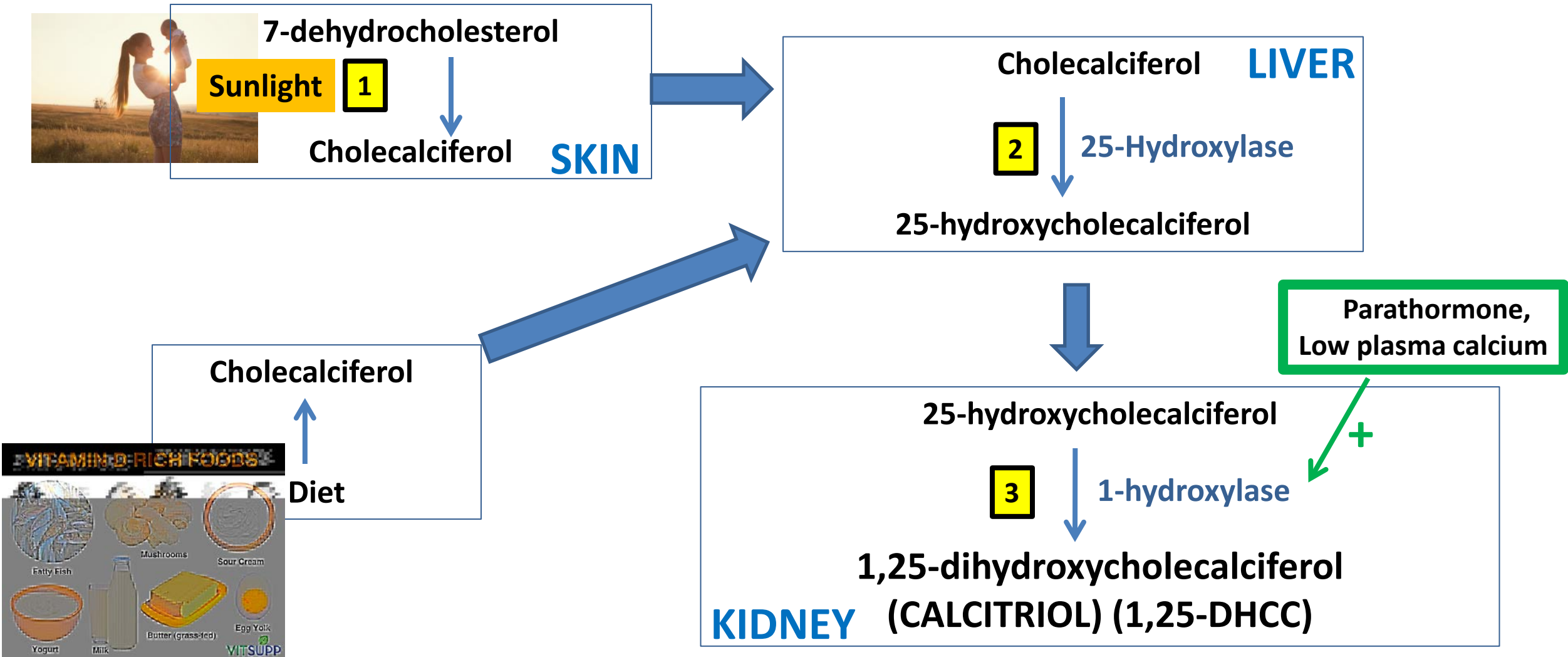
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VITAMIN D (Calciferol)



- Group of sterols that have a hormone like function
- **Ergocalciferol (vitamin D₂)** – found in plants
- **Cholecalciferol (vitamin D₃)** – Found in animal sources
- The precursor for **cholecalciferol (vitamin D₃)** synthesis in the skin is **7-dehydrocholesterol** (intermediate in cholesterol synthesis)
- Adequate exposure to sunlight can prevent vitamin D deficiency

Formation of calcitriol (1,25 dihydroxy cholecalciferol) (active form of vitamin D)



Factors affecting the formation of vitamin D in the skin

- Increased melanin pigment reduces its formation
- Time of exposure and latitude
- Amount of exposed skin/ sunscreen reduces formation
- Winter months and latitude



Inactivation of 1,25-DHCC: Action of 24-Hydroxylase

1,25-Dihydroxycholecalciferol (Calcitriol)



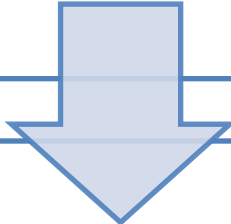
24-Hydroxylase

1,24,25-hydroxycholecalciferol (Inactive)

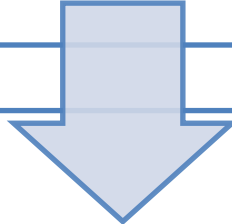
Mechanism of action of 1,25 dihydroxycholecalciferol (calcitriol)

4

1,25 Dihydroxycholecalciferol (Calcitriol) binds to intracellular receptor proteins in target cells (**intestinal mucosal cells**)



1,25-DHCC receptor complex interacts with **DNA (hormone response element) in the nucleus** of target cells (**intestine**)



Can either selectively stimulate gene expression or repress gene expression – Increases synthesis of **calbindins** (**similar to mechanism of action of steroid hormones, thyroid hormone, retinoic acid**)

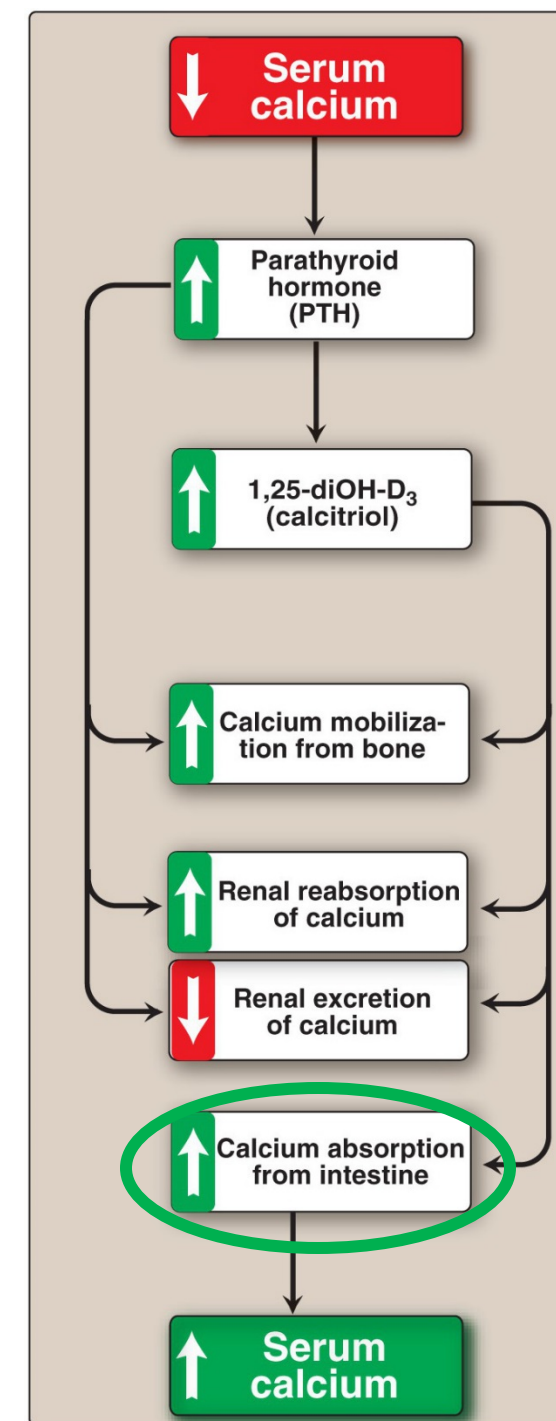
Vitamin D: Serum calcium

4

- On the **intestine** – stimulates intestinal absorption of calcium and phosphate by increased synthesis of a specific calcium binding protein (calbindin)
- On the **bone** – stimulates the mobilization of calcium and phosphate from the bone in the presence of parathormone (when the serum calcium level is low)
- On the **kidneys** – inhibits calcium excretion by stimulating parathyroid dependent calcium reabsorption

5

Increases Serum Calcium



Vitamin D: Serum phosphate

- Vitamin D **increases serum phosphate** by increasing its absorption from the intestine
- In the presence of vitamin D, serum calcium and phosphate are elevated – Increases the tendency to form bone mineral (Vitamin D facilitates mineralization, when serum calcium and phosphate are adequate)
- In vitamin D deficiency, reduced absorption of calcium reduces serum calcium level – stimulates secretion of PTH, which in turn causes demineralization of bone. Remember, PTH also increases excretion of phosphate in kidney
- **Patients with vitamin D deficiency have low serum calcium and low serum phosphate levels and decreased bone mineralization**

Vitamin D deficiency

- Risk factors:
 - Inadequate exposure to sunlight (common during winter; in elderly)
 - Nutritional deficiency – decreased intake/ fat malabsorption/ exclusively breast fed infants
 - Chronic renal disease, chronic liver disease results in decreased hydroxylation of vitamin D

Rickets: (Vitamin D deficiency in children)

- Decreased calcium absorption from the diet → Decreased serum calcium → ↑Parathyroid hormone release → ↑demineralization of bone**
- Demineralization of bone – soft pliable bones
- Characteristic **bow-leg deformity**
- Overgrowth at costochondral junction – **rachitic rosary**
- **Pigeon chest deformity**
- Frontal bossing, delayed closure of fontanelle
- Delayed teething in infants



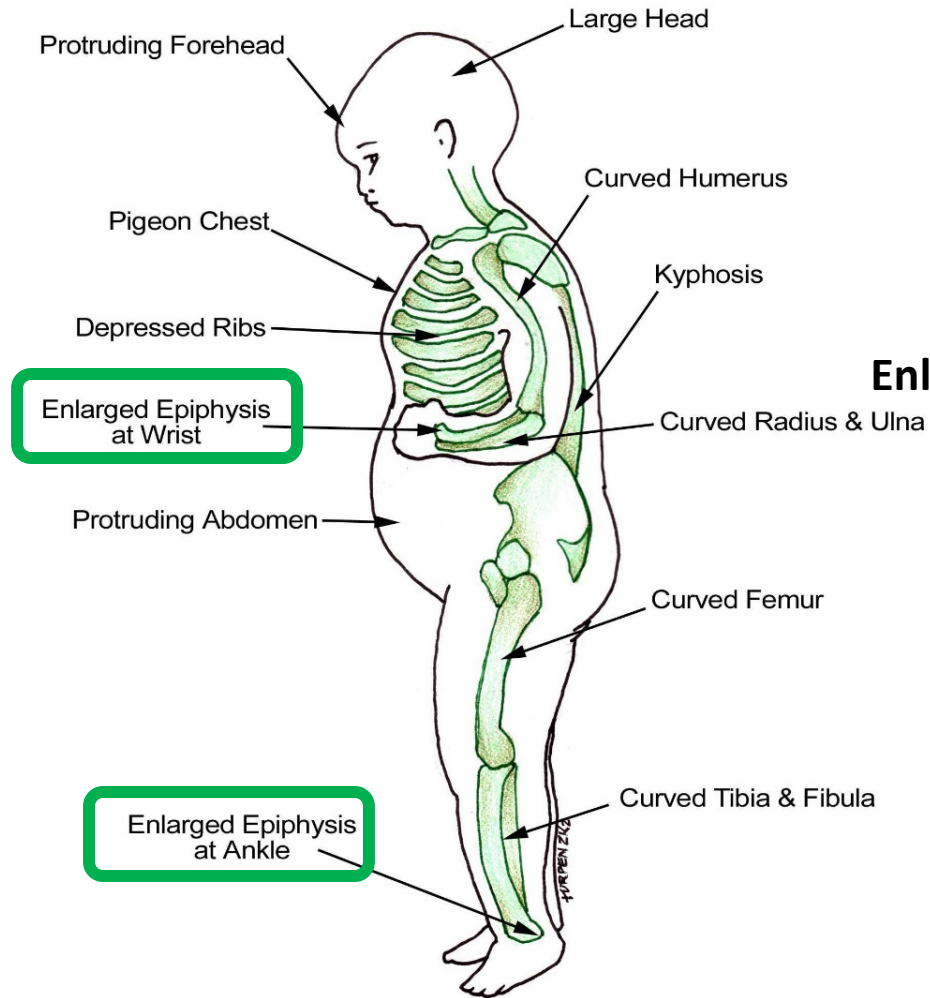
Examination of fontanelle



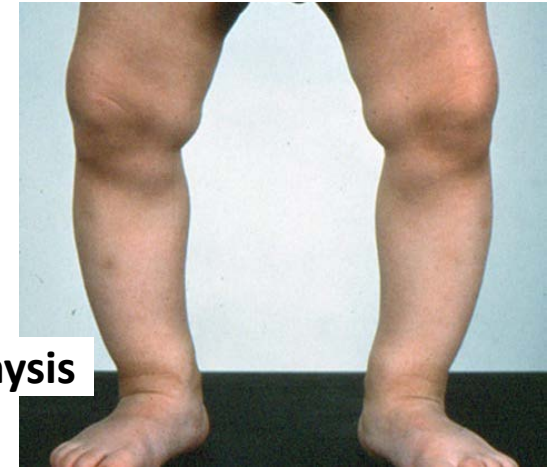
Frontal bossing



Pigeon chest deformity



Enlarged epiphysis



Bowing of legs



Bowing of legs

Medscape



Rachitic rosary

Radiologic and lab findings



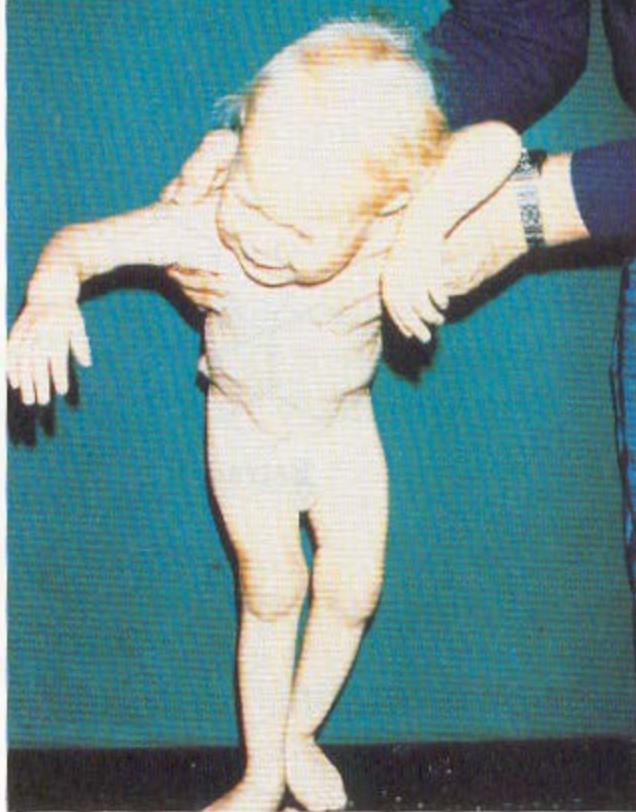
Lab findings

- Elevated **Serum alkaline phosphatase (ALP)**
- **Low serum calcium levels**
- **Low serum phosphate levels**

(A) **Rickets.** Anteroposterior radiograph of the wrist and hand in a 3-year-old child with nutritional rickets. The child had been put on a strict diet without dairy products. Note the widening, cupping, and fraying of the distal radius (arrowhead) and ulna metaphyses with an associated increase in the thickness of the growth plate (arrow). These changes are the consequence of disordered endochondral growth.

(B) **Normal.** Radiograph of the hand of a healthy 3-year-old child, without rickets.

Vitamin D and calcium supplementation



Child with rickets



After 14 months of treatment

Osteomalacia



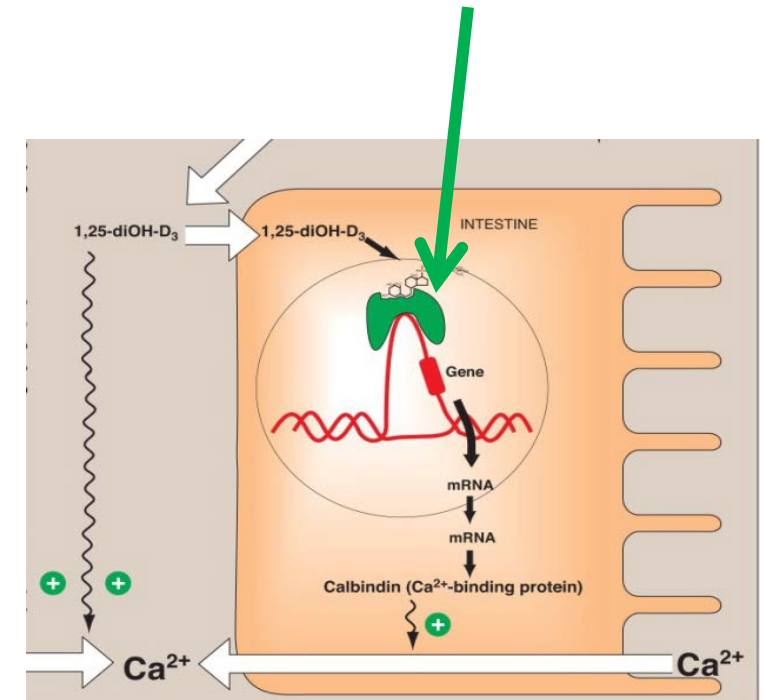
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- Osteomalacia: Vitamin D deficiency in an adult
- Bones are de-mineralized and are susceptible to fracture; May present as non-specific bone pain
- Osteomalacia can be secondary to **reduced exposure to sunlight, dietary deficiency, renal disease or liver disease**
- **Lab findings**
 - **Serum alkaline phosphatase activity is elevated**
 - **Serum calcium and phosphate levels are low**
- Subclinical vitamin D deficiency is common in the population (Teens, young adults and elderly); In teens and young adults vitamin D deficiency affects the attainment of peak bone mass

Mutations in the calcitriol receptor

- Inherited condition where calcitriol is unable to bind to its **receptor** in the intestinal mucosal cells (receptor mutation results in reduced binding of calcitriol)
- As a result, these children have high levels of calcitriol (loss of feedback inhibition)
- **Serum calcium and phosphate levels are low** but **1,25-hydroxyvitamin D levels are high**
- Compare to vitamin D deficient rickets, which has low serum calcium and phosphate and low levels of 25-hydroxyvitamin D

Calcitriol receptor mutation



Hypervitaminosis D

- Hypervitaminosis D is characterized by hypercalcemia (high serum calcium levels) – Due to increased calcium absorption from the diet
- Vitamin D also increases phosphate absorption.
- **Increase in serum calcium and phosphate** results in an increased tendency for **ectopic (soft tissues) mineralization**
- Prolonged hypervitaminosis D (prescription medications) can lead to calcification in soft tissues like kidney.

Summary

1. Formation of cholecalciferol in the skin (action of sunlight)
2. 25-hydroxylation in the liver
3. 1-hydroxylation in the kidney to form 1,25-DHCC
4. 1,25-DHCC binds to its target receptors in intestine to increase synthesis of calbindin
5. 1,25-DHCC acts on bone to increase resorption of bone

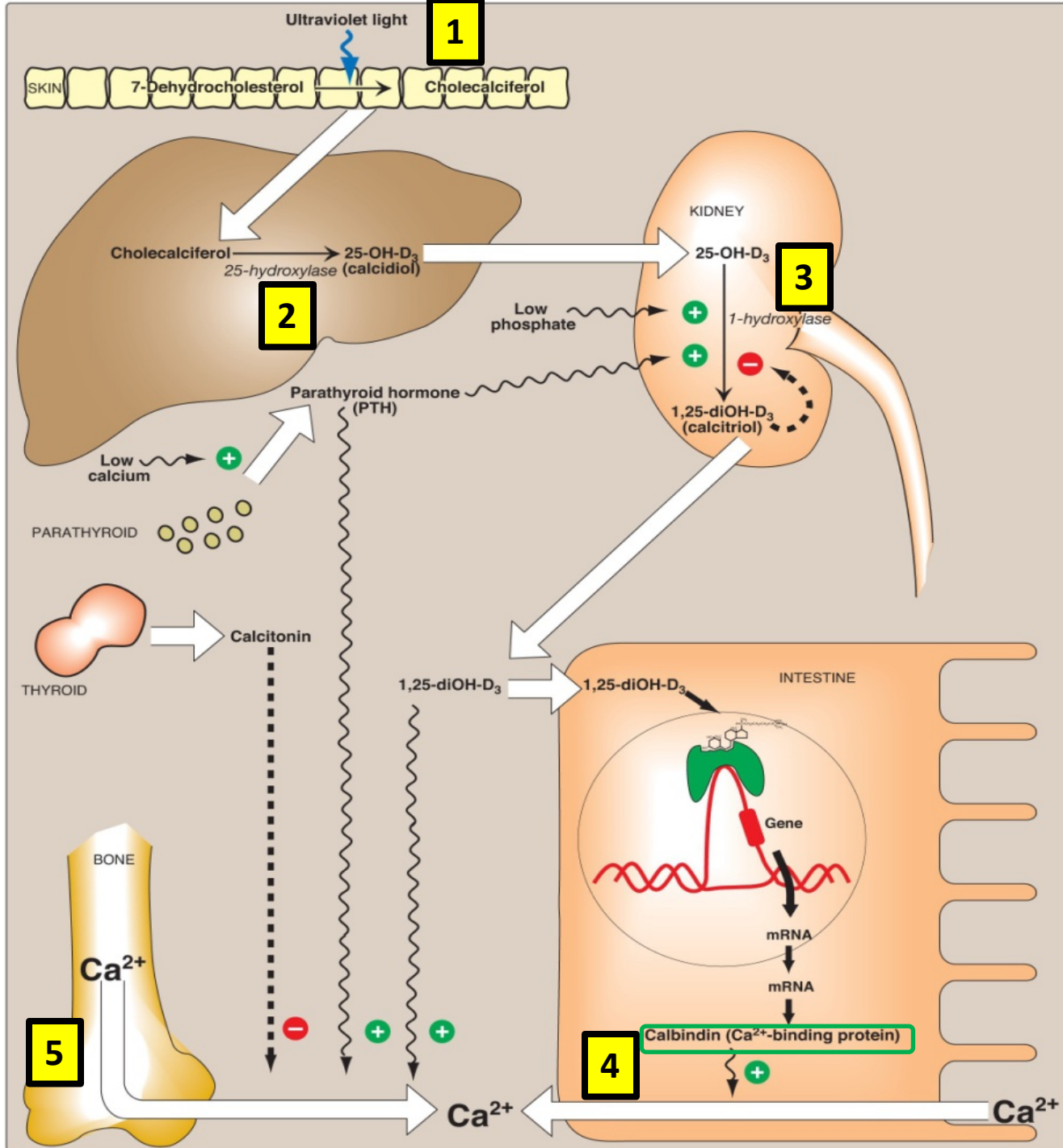


Figure 28.23

Vitamin K and E

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Vitamin K Foods

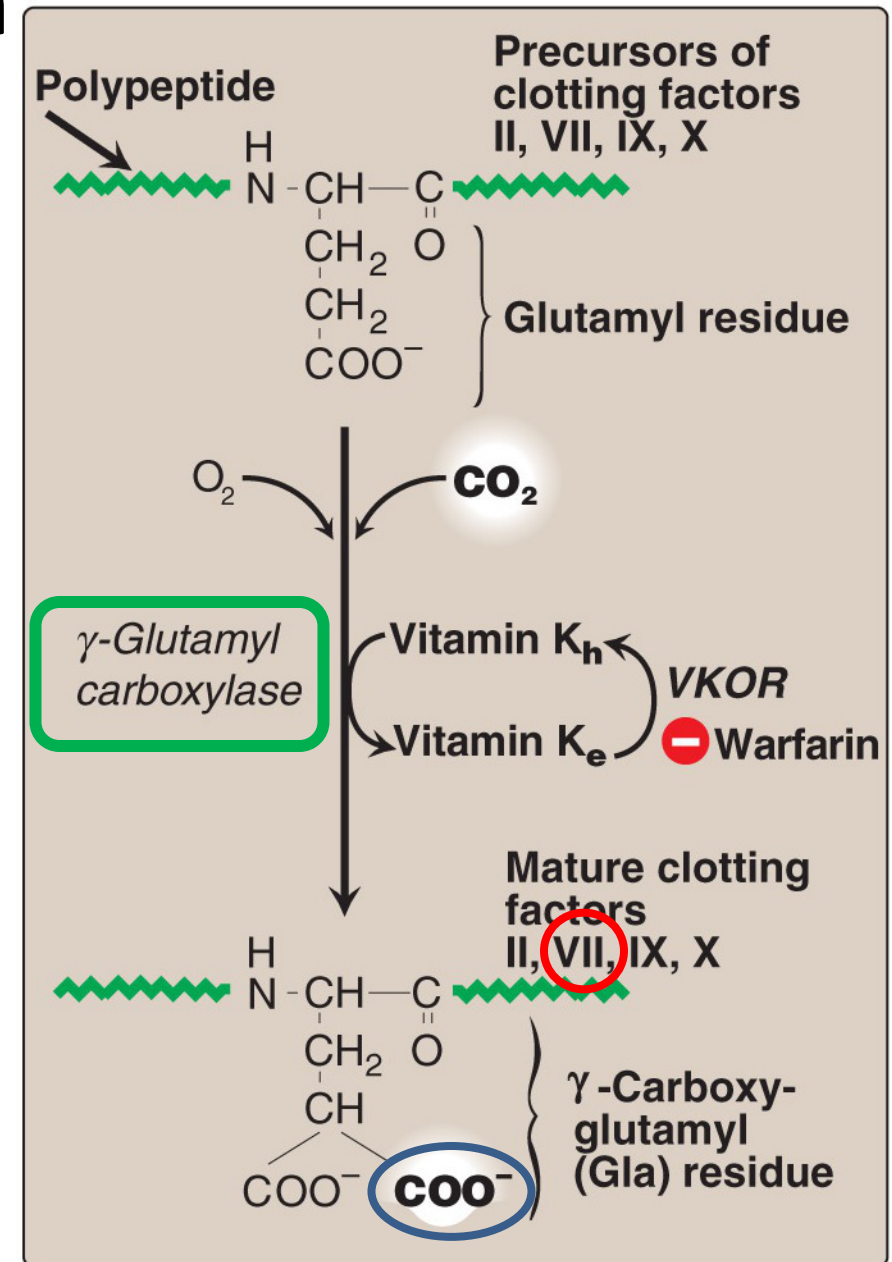
Vitamin K (Koagulation)

VITAMIN K

- Required as a coenzyme for the post translational modification (**γ -carboxylation**) of various clotting factors (**Factors II, VII, IX and X of the clotting cascade; Proteins C and S of the anticoagulant system**)
- In humans, it is also synthesized by the intestinal bacterial flora (gut microbiota) – Administration of **broad spectrum antibiotics** is a risk factor for deficiency
- Vitamin K stored in the least amounts (among fat soluble vitamins) – deficiency is the earliest to manifest
- Forms
 - Phylloquinone (plants) – dietary source
 - Menaquinone (bacteria) - intestine

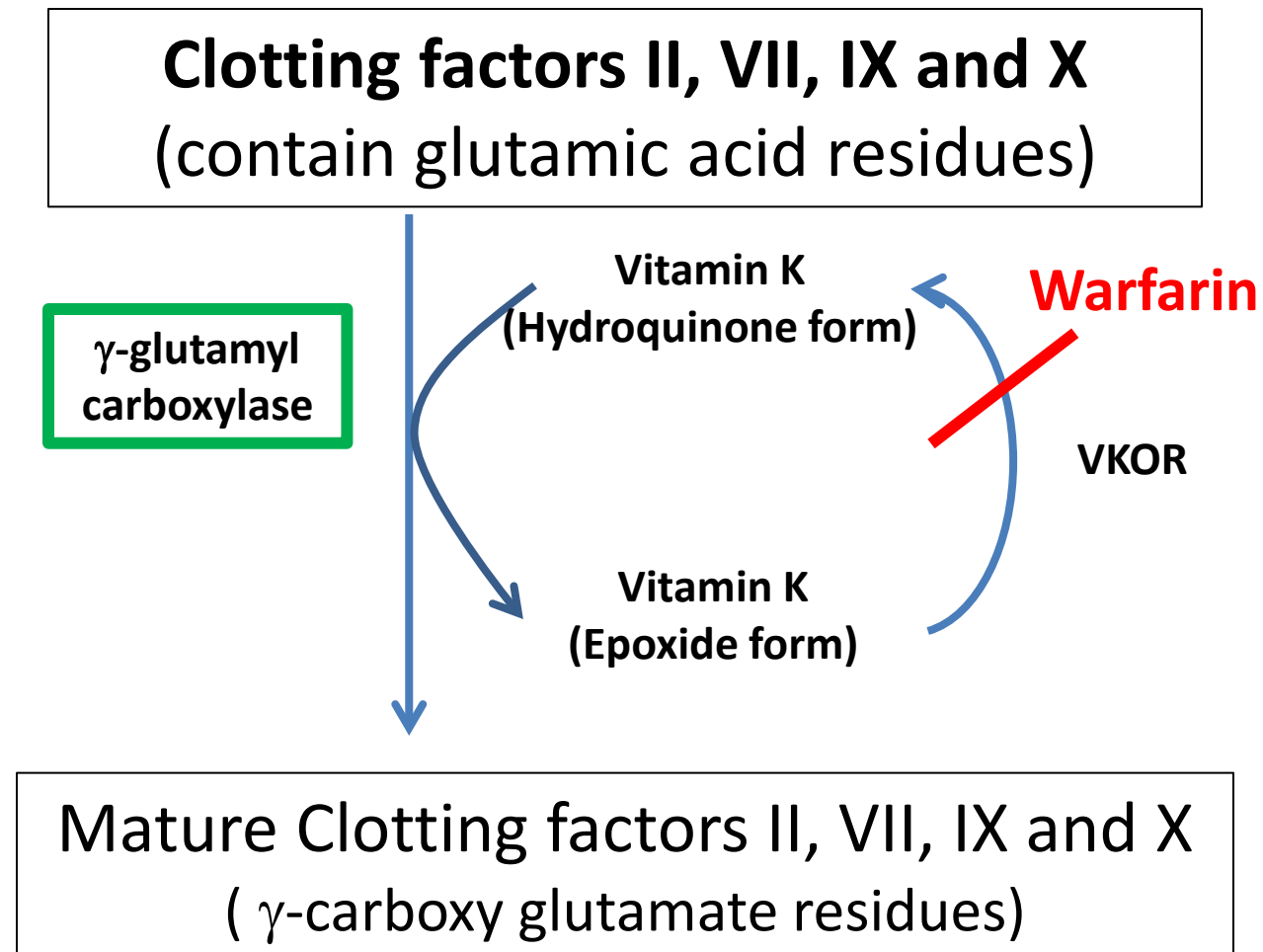
Vitamin K and gamma-carboxylation

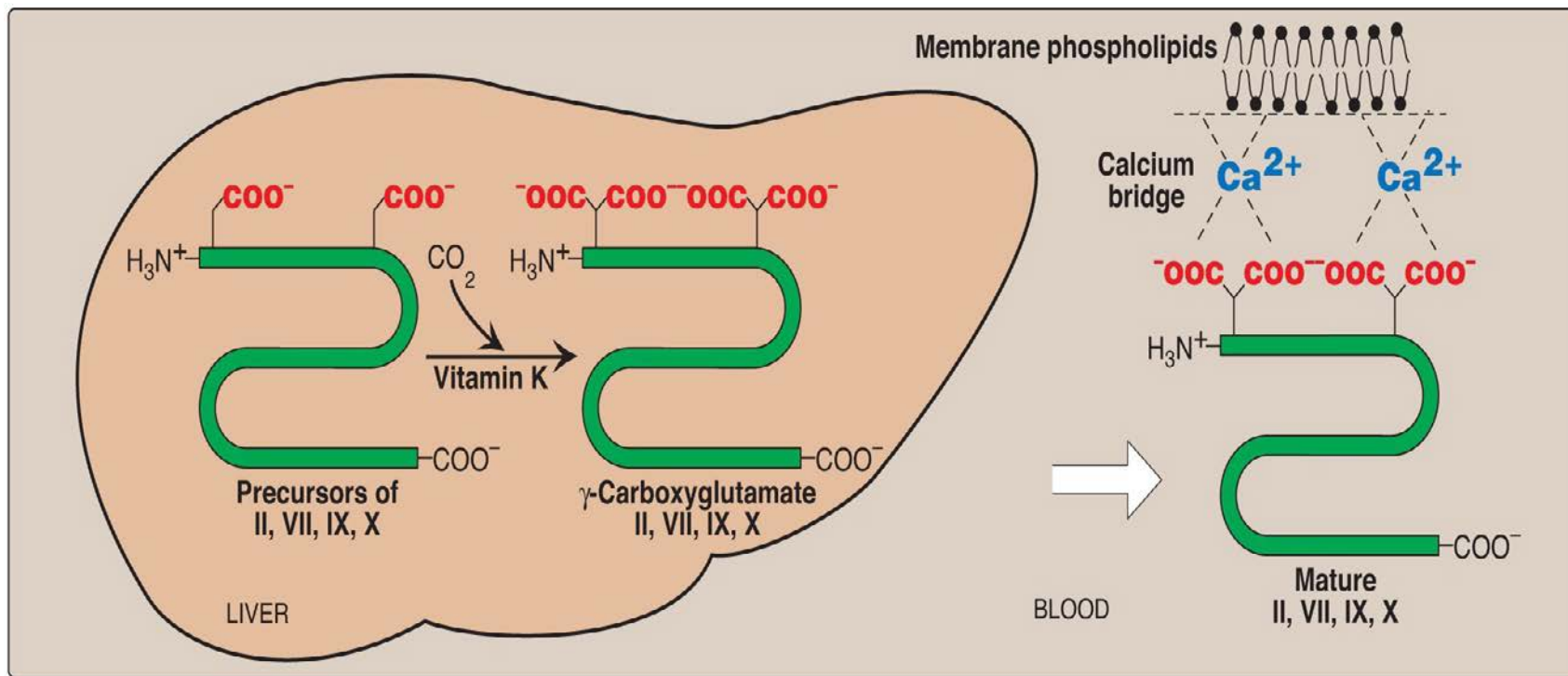
- Vitamin K is required for the hepatic synthesis of the mature clotting factors
 - **Prothrombin (II), VII, IX, X**
 - Proteins C and S (anticoagulant system)
- Vitamin K serves as a coenzyme for **liver** microsomal **γ -Glutamyl carboxylase**
- Vitamin K dependent γ - carboxylation of glutamic acid residues occurs in the liver as a post-translational modification of the clotting factors



Gamma-carboxylation and recycling of vitamin K

- γ -carboxylation of the clotting factors II, VII, IX and X occurs in the liver to form mature clotting factors (which contain γ -carboxyglutamate residues)
- Vitamin K is converted from the hydroquinone form to the epoxide form in the reaction
- **Warfarin** prevents regeneration of hydroquinone form of vitamin K – it inhibits **VKOR** (Vit K epoxide reductase)





Role of vitamin K

- γ -carboxylation (post-translational modification) forms mature clotting factors that contain γ -carboxyglutamate (Gla) and is capable of subsequent activation
- γ -carboxylation of the clotting factors facilitates **Ca^{2+} -binding** because of two adjacent negatively charged carboxylate groups
- The clotting factor calcium complex can then bind to phospholipids on the platelet membrane. (ref: Blood coagulation and fibrinolysis)

Deficiency of Vitamin K

Neonates:

- Hemorrhagic disease of the newborn
 - Bleeding at various sites in the body including skin, umbilicus and viscera
 - Intracranial bleeding – most serious complication
- Sterile intestines of the newborn – no synthesis of Vitamin K
- Routine intramuscular injection of Vitamin K for all newborns is advocated

Deficiency of Vitamin K

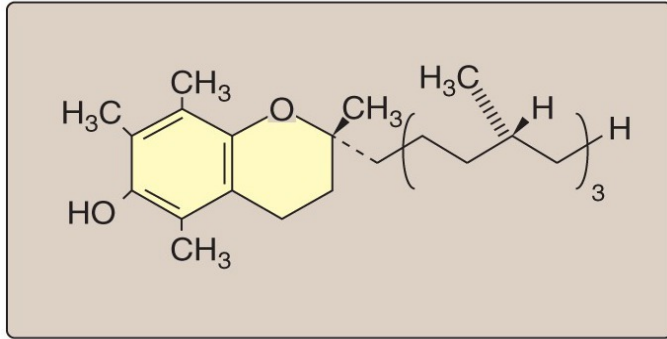
- **Risk factors:**
 - Patients with **fat malabsorption** (cystic fibrosis) and
 - Prolonged use of **broad spectrum antibiotics**
- **Laboratory tests:** Increased prothrombin time (INR)
- Bleeding disorder characterized by
 - Hematuria
 - Melena (black tarry stools)
 - Ecchymoses (bruises)
 - Bleeding from gums

Manifestations are similar to vitamin C deficiency but risk factors, biochemical basis for bleeding and lab tests are different!!!



Warfarin as an anticoagulant

- Warfarin is a vitamin K antagonist
- Warfarin blocks the activity of liver VKOR and prevent regeneration of Vitamin K – no recycling of Vitamin K – There is reduced γ -carboxylation of clotting factors and a delay in clotting
- Lab tests : Increased prothrombin time or INR (International normalized ratio)



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VITAMIN E

- Alpha tocopherol is the most active form
- Vitamin E functions as a **lipid soluble antioxidant** in cell membranes
- Due to its lipophilic nature, it can be incorporated into membranes
- **Antioxidant vitamins** are: Vitamin C (water soluble), **vitamin E and beta-carotene (provitamin A) – Lipid soluble antioxidants**
- Vitamin C and vitamin E have been shown to slow the progression of age-related macular degeneration

VITAMIN E

- Most important role of vitamin E is as a **lipid soluble anti-oxidant**
 - Prevents peroxidation of membrane lipids
 - Prevents formation of oxidized LDL (reduces risk of cardiovascular disease)
 - Scavenges free radicals generated during metabolism
- Normal diet has adequate Vitamin E, hence deficiency rare; seen mainly in patients with fat malabsorption
- Deficiency manifests as
 - **Hemolytic anemia (abnormal cellular membranes)**
 - **Reduced deep tendon reflexes and gait problems due to axonal degeneration**

Take away points

- Vitamin K – its role in the maturation of clotting factors
- Vitamin K deficiency – risk factors; manifestations and laboratory tests
- Vitamin E functions